

KT3 - Practice

Algorithmic Thinking
UE1-11 (Generics & Layered Architecture)
SS 2026

Surname(s):

Given name(s):

Student ID / Mat.-Nr.:

	maximum	achieved
Exercise 1	6	
Exercise 2	6	
Total:	12	

Time allowed: 20 minutes

This is a practice test

Instructions: See each exercise for instructions.

Question 1 (6 points) Generics: a reusable algorithm

The following interface is given:

```
interface Valued {  
    int value();    // always returns a value >= 0  
}
```

(a) [4 pts] Write a **generic method** `totalValue` that takes a `List<T>` and returns the **sum of value() over all items**.

Example: for items with values [10, 5, 25] the method returns 40.

Hint: it is a *generic method* – its header starts with `public static <T extends ...`

(b) [2 pts] Why must the type parameter be declared as `<T extends Valued>` and not just `<T>`? Explain in one or two sentences.

Question 2 (6 points) 3-Layer Architecture: where does it belong?

The method below does **everything in one place** – reading input, applying a rule, talking to the database, and updating the screen. In a clean 3-layer architecture each of these concerns lives in a different layer:

P = Presentation Layer **B** = Business Logic Layer **D** = Data Access Layer

```
public void addUser(TextField nameField) {  
    String name = nameField.getText(); // (1)  
    if (name.isBlank()) throw new IllegalStateException(); // (2)  
    String role = name.startsWith("@") ? "admin" : "user"; // (3)  
    conn.prepareStatement(  
        "INSERT INTO users(name,role) VALUES('"+name+"','"+role+"')"  
        .executeUpdate(); // (4)  
    statusLabel.setText("Added: " + name); // (5)  
}
```

(a) [5 pts] Write **P**, **B** or **D** next to each numbered marker (1)–(5) for the layer that statement belongs to.

(b) [1 pt] Name **one** concrete benefit of splitting this method into three separate layers.

